



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Department of Information and Communication Technology

ITS 4202 - Emerging Technologies

Assignment 02

Choose one freely available AI research paper or article published in the last few years (2022–2025).

You can select it from open-access websites such as arXiv.org, Google Scholar, or any university/industry blog that explains a new AI method.

Your goal is to:

1. Summarize what the research is about.
2. Identify technical or infrastructure challenges if someone tried to use it in the real world.
3. Suggest practical ways those challenges could be solved.
4. Reflect on what this means for the future of AI.

Task

1. Choose a Paper or Article (10%)

Pick a paper, article, or report related to:

- ✓ • Large AI models (e.g., ChatGPT, vision-language models)
- Edge or mobile AI
- ☐ • Energy-efficient or sustainable AI

- Privacy and federated learning
- AI safety, fairness, or robustness

2. **Summarize the Work (20%)**

In your own words, explain:

- What problem the research is solving
- What approach or method it uses
- What results or benefits it reports

3. **Analyze the Challenges (35%)**

Identify at least **two major infrastructure or technical challenges** that could appear when trying to use or scale this research.

Example topics:

- GPU or memory limitations
- Slow inference or high latency
- High energy use or cost
- Model reliability or safety problems

4. **Propose Solutions (25%)**

Suggest one or more **realistic solutions** to overcome the challenges you identified.

For example:

- Model compression or quantization
- Dynamic memory allocation
- Better hardware–software design
- Self-healing or automated retraining mechanisms

5. **Reflection (10%)**

Write a short reflection on how this research connects to the future of AI and why solving infrastructure challenges is important.

Deadline: 30th of November 2025 (11.59pm)

File format: pdf

File naming convention: ET_A2_<Index_number>.pdf